

AOUANOUK Slimane

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Available for a 6-month internship starting January 2027 in robotics, autonomous systems, or applied engineering, with an interest in data-driven and computational approaches to physical systems.

Graduate Engineering student with a strong foundation in **physics-based modeling**, **numerical simulation**, **CAD design**, and **experimental analysis** applied to **mechanical and autonomous systems**. Experienced in **CFD**, **Python-based data analysis**, and **system-level engineering**, with hands-on experience developing **AI-driven pipelines** and **physics-informed engineering systems**. Will complete a semester at the **University of Toronto** (Fall 2027) with coursework in **Data Modeling**, **Machine Learning**, **Deep Learning**, and **Data-Driven Decision Systems** to further strengthen a **data-driven engineering** profile.

EDUCATION — **Combined GPA : 3.92 / 4.00**

Arts et Métiers Institute of Technology (ENSAM) — Paris, France **Sept 2024 – Jul 2027**
Combined BS × MS Engineering Degree — Arts et Métiers is a member of **ParisTech**, a consortium of prestigious French institutions recognized for **academic excellence**, **outstanding faculty**, and **world-class research laboratories**.

- **Ranked top 13% of cohort** (160 / 1203 students)
- **Relevant Coursework:** Mechanical Design and Structural Analysis, Solid Mechanics, Robotics and Mechatronics, Computer-Aided Design (CAD), Design for Manufacturing and Assembly (DFM/DFA), Engineering Materials, Experimental Mechanics and Validation Testing, Fluid Mechanics, Heat Transfer, Numerical Optimization

SELECTED PROJECTS

Autonomous Trajectory Optimization (PIMM Lab) — Supervised by Prof. Francisco Chinesta **2026**

- Research project on drone trajectory optimization: developed a **Python-based framework** using **Euler–Lagrange** variational formalism to compute **energy-aware flight trajectories**; integrated **dynamic cost modeling** (velocity, acceleration, wind effects) and **obstacle avoidance**; progressing toward a **digital twin** approach using **Physics-Informed Neural Networks** for **real-time decision-making**; research-oriented project with **publication potential**.

Miniature Formula One Car Design – Arts et Métiers **2025**

- Designed a miniature F1 car optimized for a 20 m straight-line race; developed the full CAD model in **Fusion 360** with 3–4 design iterations; conducted **CFD simulations** in **STAR-CCM+** at 50 m/s to analyze pressure and drag; used simulation results to guide aerodynamic optimization; produced a 3D-printed prototype and prepared CNC machining while ensuring compliance with 10+ technical regulations

AI-Driven Smart Workshop Project – Arts et Métiers **2025**

- Built an **end-to-end local data & AI pipeline** to query **technical workshop data** in **natural language**; structured **heterogeneous CSV data** and migrated it to a **relational SQLite database**; **validated and explored data** using **Python** and **pandas**; designed **SQL views** to simplify **complex joins** and improve **robustness**; implemented a secure **Text-to-SQL system** powered by a **locally deployed LLM (Ollama)** with **strict query constraints**; developed a **Streamlit interface** enabling **real-time usage** and **human-readable responses**

Aircraft Brake System Design Project – Arts et Métiers **2025**

- Designed a **complete mechanical braking system** for a **light aircraft wheel** composed of **10+ components** using **Fusion 360**; built a **physics-based functional architecture** ensuring **torque transmission** across **5+ mechanical interfaces**; **quantitatively modeled load paths** and **mechanical constraints**; performed **engineering validations** including **bearing preload analysis**, **bolt slip criteria**, **shaft stress verification**, and **brake disc thermal dissipation**; ensured **system consistency** through **analytical checks** and **safety margins**

WORK EXPERIENCE

Construction Site Worker Intern – Société de Rénovation Parisienne (SRP) **Summer 2025**

- Supported large-scale construction operations on a major renovation project involving **3,000+ tons of excavated soil**. Conducted **on-site measurements, markings, and consistency checks** under real-world tolerances during a **9–10 month underground phase**. Coordinated with subcontractors and adapted plans to field constraints, gaining hands-on exposure to **field data**, safety requirements, and execution–theory gaps

SKILLS AND INTERESTS

Data & Programming: Python (data analysis, numerical computation), SQLite, MATLAB, Arduino, LaTeX

Modeling & Simulation: Computational Fluid Dynamics (STAR-CCM+), Finite Element Analysis (Abaqus), Mechanical CAD (Fusion 360, 3DEXPERIENCE), Simulink

Experimental Tools: Video-based motion analysis, sensor-based measurements

Languages: French (native), English (fluent; TOEFL iBT: 88/120), Italian (basic), Spanish (basic)

Interests: Data-driven engineering, applied AI, technology ecosystems (Apple), basketball, soccer, strength training